ASML Holding N.V. (ASML)

The EUV Cycle Pullback Masks a Multi-Year Moat Expansion

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> ASML's recent earnings miss reflects a temporary slowdown in EUV tool shipments rather than a structural erosion of its market dominance. While consensus models over-price the near-term revenue dip, we see a durable pricing power backed by a near-monopoly in high-NA lithography and a pipeline of next-generation platforms. Our blended target of $1,442.46 implies a 12.1% downside versus the current $1,641.52 price, reinforcing a Hold with valuation concerns.

HOLD	$1,442.46	-12.1%	$1,641.52
Hold, Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The market fixates on the quarterly dip in EUV orders, yet the compounding effect of ASML's technology lock-in will drive earnings growth over the next decade.

EXECUTIVE SUMMARY

ASML operates the only commercially viable extreme ultraviolet (EUV) lithography platform, supplying the most advanced nodes to a handful of integrated device manufacturers. In 2025 the company captured roughly 70% of the EUV market, with a revenue base exceeding $23.5 billion and a free cash flow generation of $6.0 billion. The firm's product mix is heavily weighted toward high-NA EUV systems, which command premium pricing and generate recurring service revenue. The recent earnings shortfall stems from a modest slowdown in high-NA shipments, a phenomenon that analysts have amplified into a broader growth narrative. The catalyst for the current multiple compression is the market's reaction to a single quarter of lower tool acceptance, combined with heightened macro uncertainty in the semiconductor cycle. This over-reaction has pulled ASML's price-to-sales multiple below its historical mean, creating a valuation gap that we believe will close as the high-NA pipeline matures and demand from emerging applications such as advanced packaging accelerates. Capital allocation remains disciplined. ASML continues to invest roughly 15% of revenue into R&D, targeting the next generation of high-NA and extreme-high-NA (E-HX) systems. The balance sheet is strong, with net cash of $12 billion and a leverage ratio well below 0.5x. No material M&A activity is anticipated, allowing the firm to focus on organic innovation and incremental capacity expansion.

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OPERATIONAL SNAPSHOT

$1,641.52	$23.5 B	$6.0 B	26%	1.30	$1,641.52
REF. PRICE	LTM REVENUE	LTM FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

LTM figures are Cascadia estimates grounded in the latest filings; ref. price as of 2026-10-16.

THE MOAT IS FORMIDABLE AND DURABLE

ASML's moat rests on a combination of switching costs, regulatory barriers, and a deep patent portfolio that together create a near-monopoly in high-NA EUV lithography. Customers face multi-year integration cycles, extensive training requirements, and a supply chain that is tightly coupled to ASML's tool architecture, making any switch to a hypothetical competitor prohibitively expensive. Regulatory scrutiny adds another layer of protection. The EU's export controls on advanced lithography equipment limit the number of qualified suppliers, and ASML's compliance infrastructure has been honed over a decade of navigating these constraints. This environment discourages new entrants and reinforces the company's pricing power. Patents covering the core optics, light source, and mask handling systems underpin a defensible technology edge. ASML files over 300 patents annually, and its portfolio is cited by rivals as a barrier to entry. The firm's ability to monetize this IP through licensing and service contracts further entrenches its cash flow stability. Customer retention is exceptionally high. The three major integrated device manufacturers collectively account for over 90% of ASML's revenue, yet each has committed to multi-year roadmaps that lock in tool purchases for the next 10 to 15 years. This alignment of product development cycles ensures a predictable revenue stream and limits price erosion.

VALUATION

Three scenarios make the valuation range explicit

The baseline case assumes an 8% compound annual revenue growth rate, a gradual improvement in free cash flow margin to 26% as high-NA capacity ramps, and a terminal growth of 3% reflecting the long-run expansion of the semiconductor ecosystem. The bear case trims revenue growth to 5% and compresses margins to 22% amid a prolonged slowdown in advanced node adoption. The bull case envisions a 12% revenue CAGR driven by accelerated adoption of high-NA in emerging applications and a margin expansion to 30% as economies of scale materialize.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	5%	22%	$1,160.65	-29.3%
Base	50%	8%	26%	$1,414.67	-13.8%
Bull	25%	12%	30%	$1,779.85	+8.4%

Bear $1,160.65

The bear outcome materializes if high-NA shipments falter for two consecutive years, if macro-level demand for advanced nodes contracts, and if competitive pressure from alternative patterning technologies intensifies.

Base $1,414.67

The base case reflects a steady rollout of high-NA tools, modest growth in advanced packaging demand, and continued service revenue expansion, all of which support an 8% revenue trajectory and stable margins.

Bull $1,779.85

The bull scenario requires an early breakthrough in extreme-high-NA lithography, accelerated adoption of 2-nm and sub-2-nm nodes, and a favorable regulatory environment that limits any potential new entrant.

VALUATION FRAMEWORK

A live discount rate, not a static hurdle

Our cost of capital is derived from a CAPM that incorporates the 1.30 equity beta, a risk-free rate of 4.0%, and an equity risk premium of 6.0%, resulting in a WACC of 10.00% as of 2026-10-16. This live discount rate captures the market's current assessment of ASML's systematic risk and ensures that the valuation reflects real-time pricing dynamics rather than a static hurdle. The 10% WACC also signals that investors demand a premium for the technology-intensive nature of the business, yet the premium is modest given the firm's entrenched market position. Consequently, the discount rate does not overly penalize the projected cash flows, allowing the intrinsic value to remain attractive relative to the current market price.

FCF margin normalization is the engine

Across the three scenarios we model a trajectory in which free cash flow margins converge toward a long-run equilibrium of 30% as high-NA capacity utilization improves and service contracts mature. In the base case the margin stabilizes at 26% by year five, reflecting incremental cost efficiencies and a higher proportion of recurring revenue. The bear case assumes a slower margin accretion due to under-utilized capacity and higher R&D spend relative to revenue, while the bull case accelerates margin expansion as the firm leverages its scale to negotiate better component pricing and reduces per-unit amortization of capitalized development costs.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A and stock compensation, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

KEY VALUATION PARAMETERS

Revenue base year	**2026**
Forecast horizon	**10 years**
Live WACC	**10.00% (2026-10-16)**
Terminal growth	**3.0%**
Equity beta	**1.30**
Diluted shares	**1.0 bn**

HOLD

over $1,442.46 blended target

Methodology. Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

BUSINESS & SEGMENT DRIVERS

Revenue prospects anchor on high-NA EUV adoption and advanced packaging demand

HOLD

High-NA EUV shipments remain the primary growth engine, with each system delivering roughly $150 million in revenue over its lifecycle. The rollout schedule for the 0.55 NA platform is expected to add $3.5 billion of revenue annually by 2029, driven by the three major chipmakers' transition to 2-nm and beyond. A secondary driver is the surge in advanced packaging, where ASML's lithography-plus-metrology solutions command premium pricing and generate recurring service contracts that lift the overall margin profile. The DUV segment, while mature, continues to provide a stable cash flow base. DUV tools account for approximately 30% of total revenue and benefit from incremental upgrades that extend the useful life of existing equipment. Service and spares revenue, which represents roughly 20% of total sales, adds a high-margin, low-volatility component to the cash flow mix. We anticipate that the convergence of high-NA and advanced packaging will create cross-selling opportunities, allowing ASML to bundle lithography with metrology and inspection solutions. This integrated offering not only deepens customer lock-in but also expands the addressable market beyond traditional logic nodes.

What to monitor

- **High-NA order backlog** A rising backlog signals sustained demand and validates the revenue trajectory projected in the base case.
- **Advanced packaging adoption rates** Faster uptake of chip-on-wafer and fan-out technologies could accelerate service revenue growth.
- **Regulatory export approvals** Any shift in EU or US export policy could affect the timing of tool deliveries to key customers.
- **R&D execution on E-HX** Successful milestones on extreme-high-NA development would underpin the bull scenario.

70%	$150 M	26%	$12 B	15%	1.30
EUV MARKET SHARE	REVENUE PER HIGH-NA TOOL	FREE CASH FLOW MARGIN	NET CASH	R&D INTENSITY	EQUITY BETA

METRIC	FY 2025
Revenue	$23.5 B

Gross Profit	$15.0 B
EBIT	$7.0 B
Net Income	$5.5 B
Free Cash Flow	$6.0 B
R&D	$3.5 B
ROE	28%

RISK & MOAT MATRIX

The bear case

SHARE EROSION

A prolonged slowdown in high-NA adoption could erode market share if a competitor emerges with a viable alternative, leading to a revenue contraction.

M&A DIGESTION

Any future acquisition aimed at expanding the metrology portfolio may strain cash flow if integration costs exceed synergies.

REGULATORY AND PRICING SCRUTINY

Heightened export controls or antitrust investigations could delay shipments and compress pricing power.

TECHNOLOGY DISRUPTION

Breakthroughs in directed self-assembly or alternative patterning could reduce the long-term relevance of EUV.

REINVESTMENT TREADMILL

Escalating capital intensity required to sustain the high-NA roadmap may pressure free cash flow if revenue growth stalls.

WACC AND DURATION SENSITIVITY

An increase in market volatility could lift the discount rate, reducing present value of long-term cash flows.

The strengths

OLIGOPOLY POSITION

Control of the EUV supply chain creates a structural barrier that limits competitive encroachment and sustains pricing power.

FCF COMPOUNDING

Recurring service contracts and high-margin tool sales generate a virtuous cycle of cash generation that funds ongoing innovation.

GEOGRAPHIC DURABILITY

Global customer footprint across the United States, Europe, Taiwan, and South Korea insulates the business from regional downturns.

PIPELINE OPTIONALITY

The parallel development of high-NA, extreme-high-NA, and advanced packaging solutions provides multiple growth levers.

Compelling risk-reward, skewed to the upside

We conclude that ASML’s entrenched market position, disciplined capital allocation, and robust cash flow generation outweigh the near-term earnings volatility. The valuation gap created by the recent multiple compression offers a modest entry point, yet the upside potential remains limited by the already high price relative to fundamentals. Accordingly, we maintain a Hold with valuation concerns, a blended target of $1,442.46 and an implied 12.1% downside versus the current $1,641.52 price.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

FRANCHISE DEEP DIVE

High-NA EUV: repriced on normalization

High-NA EUV represents the flagship product line that underpins ASML’s premium pricing. The platform delivers a 0.55 numerical aperture, enabling patterning at sub-2-nm nodes with a throughput advantage of roughly 30% over the previous generation. In the last twelve months the market re-rated the segment downward as customers delayed purchases amid macro uncertainty, compressing the price-to-sales multiple from 12× to 9×. The normalization narrative rests on two pillars. First, the technology has reached a production-ready state, and the three major chipmakers have publicly committed to a multi-year high-NA rollout. Second, the service and spares component of the high-NA ecosystem has matured, delivering a recurring revenue stream that cushions the impact of any short-term order lag.

Why the growth numbers disappointed

- Customers postponed high-NA orders to align with capital-intensive fab expansions.
- Supply-chain bottlenecks in laser source components limited initial delivery capacity.
- Analyst models over-estimated the speed of transition from 3-nm to 2-nm nodes.
- Macro-level semiconductor demand softness reduced the urgency of tool upgrades.

Our read: Normalization, not share loss, defines the near-term outlook. The market's over-reaction to a single quarter of slower shipments has created a valuation discrepancy that we expect to resolve as high-NA adoption accelerates and service revenue expands.

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